ALICE L WALTON SCHOOL OF MEDICINE

Retroviridae (Part 2 of 2)

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Explain disease progression and epidemiology of HIV.
2. Describe the life cycle of HIV.
3. Interpret laboratory results to identify HIV.

CASE CONNECTION

A 34-year-old male presents to the clinic with painless, white, corrugated lesions on the lateral border several weeks. The patient reports that t he tries to remove them vigorously with a toothbrush. He denies any recent antibiotic use. Physical examination reveals the characteristic "hairy" appearance of the lesions.

What is the most likely etiological agent responsible for these oral lesions? Aside from the oral lesions, are there other concerns?

GO TO CONCLUSION ↓

Why is There No Cure for HIV? (LO3)

So, if we know so much about the replication cycle of HIV, why is there no cure?

Two key steps in the HIV replication pathway are reasons why a cure for HIV has been so challenging.

Reverse Transcriptase: No Proofreading

Reverse transcriptase does not have the ability to proofread when viral DNA is being made from viral RNA. Proofreading in replication is like proofreading in writing. As you write, you make mistakes. Recognizing those mistakes and correcting them is essential to producing a coherent, final message. Otherwise, no one woudl hae any ieda waht yuo weer triyng ot sya.

When making copies of RNA or DNA from a DNA template, mistakes will be made. Keeping those mistakes, especially in daughter cells, could be deadly in humans and other eukaryotes. Repeated mutations in humans carried to daughter cells is a key reason we get cancer. Your cells make mistakes, those mistakes compound passed down to daughter cells, and eventually, cancer occurs.

So how does a lack of proofreading help HIV? Most of the time it doesn't. When reverse transcriptase makes viral DNA from RNA, the mutation actually impedes the virus. While some mutations are helpful, most are detrimental to the virus.

However, HIV makes billions of copies of itself in a single day inside the body. Even if most of those mutations made are problematic, a few, by random chance, will benefit the virus. Those mutant viruses will spread throughout the host and become the dominant strain. The overall effect is that HIV can rapidly adapt to the host immune response and become resistant to antiviral therapy.

CLINICAL CORRELATION

The first drug approved by the FDA to treat HIV was zidovudine, also known as AZT. AZT alone was not successful in preventing HIV patients progressing to AIDS. The high mutation rate of HIV allowed it to quickly adapt to AZT and become resistant to treatment.

Today, we use combinations of drugs to treat HIV, making it much more difficult for the virus to mutate around treatment plans.

Incorporation of Viral Genome into Host DNA

The second reason HIV is so difficult to cure is that once reverse transcriptase coverts viral RNA to viral DNA, the viral DNA is incorporated into the host genome. This step is ell's life, the virus DNA will be inside its genome. Every daughter cell that is created from that cell will also have the viral DNA copied into its genome. In order to cure HIV from a patient, every cell that has HIV DNA in its genome would have to be destroyed or have the viral DNA cut back out.

This is why ART can suppress viral titers in the blood below the limit of detection without curing the infection. The virus is still in the patient's DNA, even if you cannot detect it in the blood anymore. If a patient stops taking ART, viral titers will rapidly begin to rise in the blood again.

While there is no practical cure for HIV, an HIV+ patient on ART therapy has roughly the same life expectancy of a patient without HIV. Again, this highlights the need to get patients screened for HIV for their health and to minimize the risk of spreading the virus.

CLINICAL CORRELATION

Notice that I said, "practical cure for HIV?" 5 people have been cured of HIV to date. The first patient was Timothy Ray Brown, also known as the Berlin patient. Ray was diagnosed with HIV and later developed leukemia. A bone marrow transplant was preformed to treat his cancer, but the donor stem cells had a mutation in CCR5 (CCR5Δ32 homozygous). Because of this mutation, HIV could not get into cells efficiently, and Ray was cured. The other patients who were cured were given similar bone marrow transplants with the same mutation in donor cells.

There are some people who have naturally occurring mutations in CCR5 that make them very resistant to HIV infections. However, the virus is crafty and can still infect these patients if it binds the CXCR4 coreceptor!

How does an HIV infection progress, and what are the symptoms? (LO2)

Once contracted, an HIV infection occurs in 3 phases:

- **Acute phase**
- **Chronic phase**
- **AIDS (Crisis)**

Acute phase: After exposure, HIV will replicate in local tissue antigen-presenting cells and CD4 cells. Shortly after, the infection will spread to local lymph nodes and rapid replication of the virus will begin. The virus spreads via the blood to distal sites and seeds in the lymphoid organs systemically. This causes an acute viremia in patient and high viral titers in the blood. CD4 T cell count may fall significantly, but rarely <400 CD4 T cells/uL.

CLINICAL CORRELATION

The acute viremia during the acute phase can present as a mononucleosis-like illness in patients that is clinically indistinct from EBV or CMV. This typically presents 3-6 weeks after the infection and symptoms usually resolve in 2-4 weeks. Note that not all patients have symptoms during the acute phase of HIV.

Roughly 40-90% of patients will have a mononucleosis-illness within the first few weeks of infection. Some patients have no signs of infection at all.

Chronic phase: Several weeks after infection, patients begin to produce antibodies against HIV surface antigens. This binds free floating virions and targets them for destruction by phagocytes. In addition, cytotoxic T cells numbers increase and begin hunting down and destroying infected cells. This causes viral titers to decrease, and the patient's symptoms, if present, subside.

However, the immune system cannot clear the virus completely. Despite the patient having no symptoms, the virus is actively replicating as the immune system fights the infection. The chronic phase can last a decade or more.

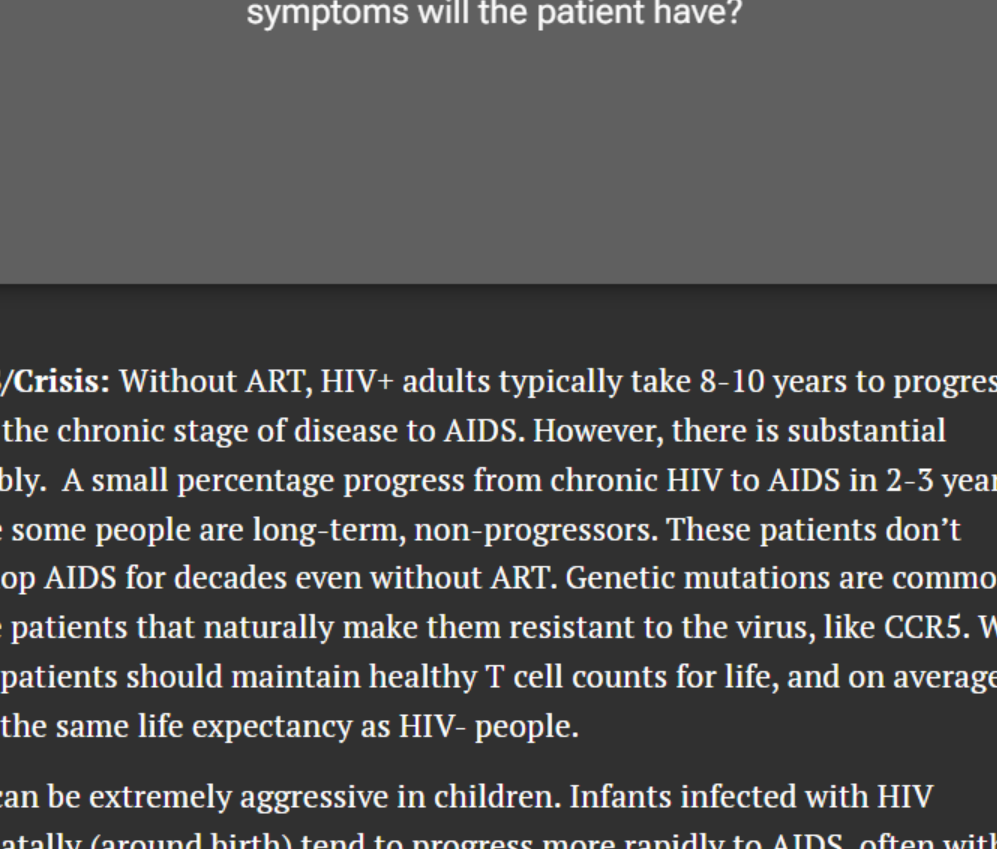


Figure 9: CD4 T cells counts and viremia in an HIV patient over time.

During the acute phase of HIV, if symptoms are present, what types of symptoms will the patient have?

AIDS/Crisis: Without ART, HIV+ adults typically take 8-10 years to progress from the chronic stage of disease to AIDS. However, there is substantial variably. A small percentage progress from chronic HIV to AIDS in 2-3 years, while some people are long-term, non-progressors. These patients don't develop AIDS for decades even without ART. Genetic mutations are common in these patients that naturally make them resistant to the virus, like CCR5. With ART, patients should maintain healthy T cell counts for life, and on average, have the same life expectancy as HIV- people.

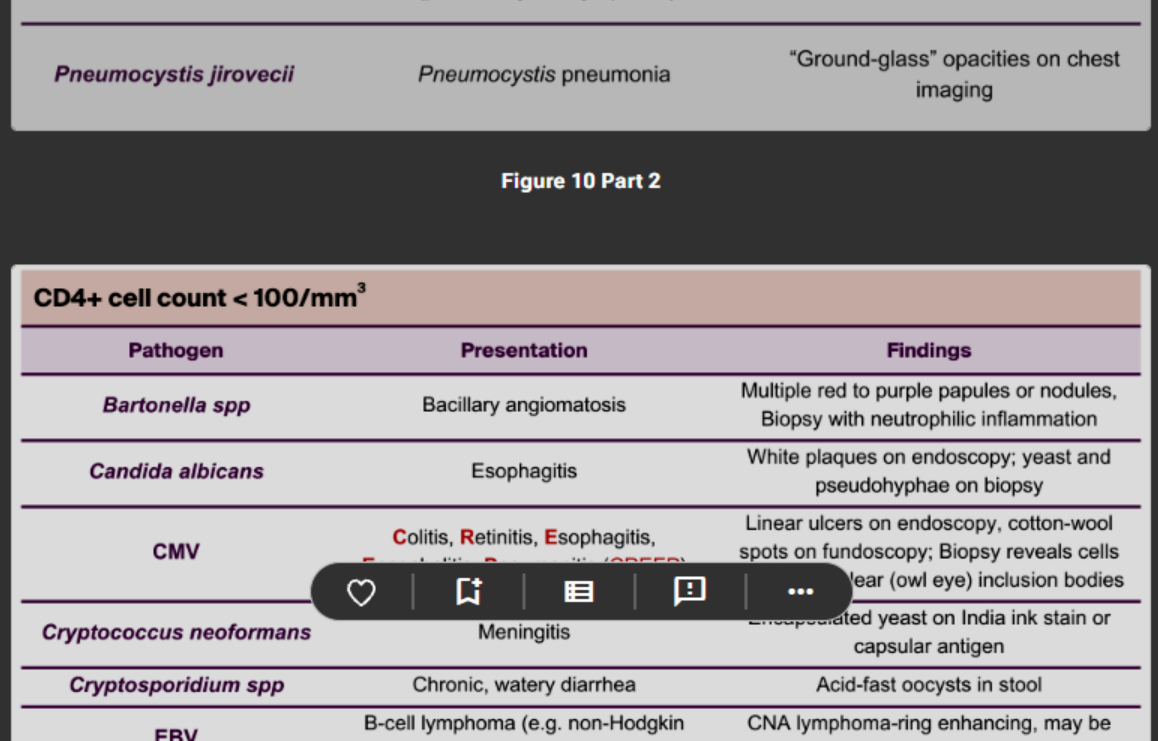
HIV can be extremely aggressive in children. Infants infected with HIV perinatally (around birth) tend to progress more rapidly to AIDS, often within 1 to 2 years, if not treated.

AIDS Criteria: There AIDS (6 years to adults).

1. T cell count <200/uL (Normal Range: 500-1500/uL)
2. Any patient with HIV and diagnosed with an infection or cancer indicating a severe immunodeficiency. These are referred to as AIDS-defining condition.

Once a patient falls below a CD4 T cell count of 200/uL, they are severely immunosuppressed and at risk for serious, lethal infections and cancers. Note that some infections commonly associated with HIV can occur at CD4 T cell levels higher than 200/uL. As CD4 T cell counts drop, the risk of reactivation of latent viruses, like herpesviruses, increases. These infections can be severe, unlike anything you would see in a patient with a heathy immune system.

Here's a list of infections commonly seen in HIV+ patients at specific T cell counts. Please note these are not magical numbers-it is possible for patients to get these infections at higher or low T cell counts. This table just represents the most common T cell counts where these illnesses are seen.

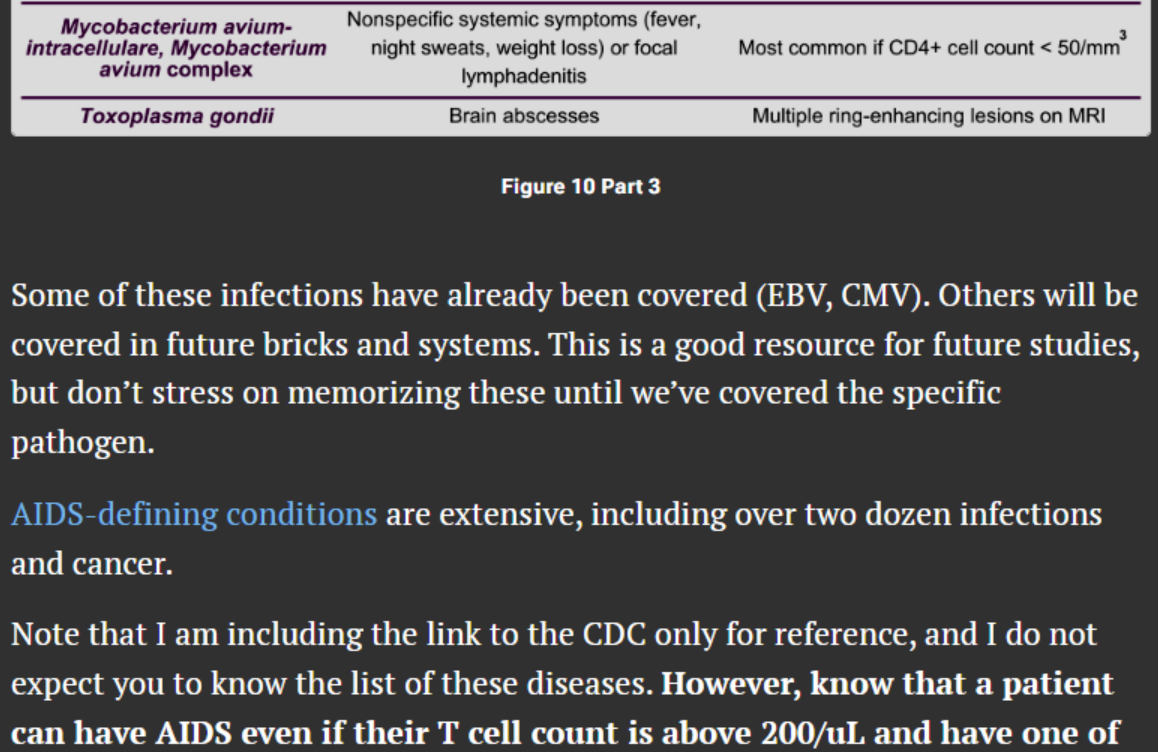


COMMON DISEASES OF HIV-POSITIVE ADULTS ↓CD4+ cell count ⟶ reactivation of past infections (e.g. TB, HSV, shingles), dissemination of bacterial and fungal infections (e.g. coccidioidomycosis), and non-Hodgkin lymphomas.

CD4+ cell count < 500/mm³

Pathogen	Presentation	Findings
Candida albicans	Oral thrush	Scrapable white plaque, pseudohyphae on microscopy
EBV	Oral hairy leukoplakia	Unscrapable white plaque on lateral tongue
HHV-8	Kaposi sarcoma, localized cutaneous disease	Perivascular spindle cells invading and forming vascular tumors on histology
HPV	Squamous cell carcinoma at site of sexual oropharynx)	
Mycobacterium tuberculosis	Increased risk of reactivation of latent TB infection	

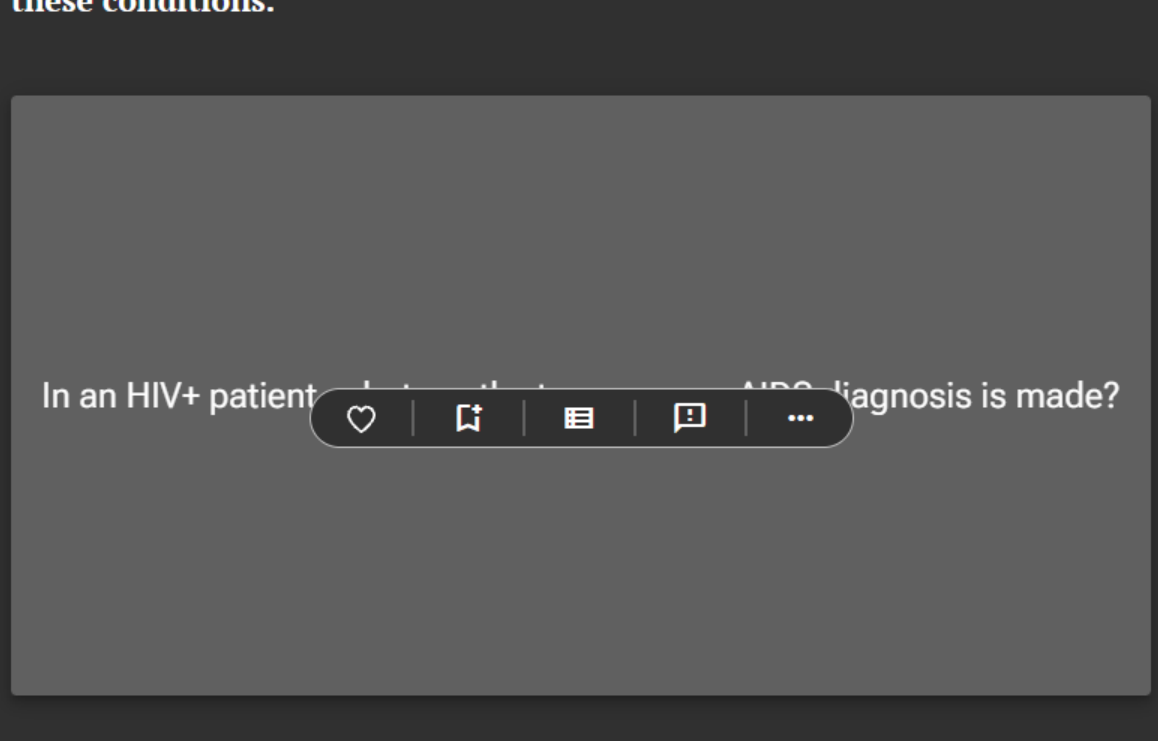
Figure 10 Part 1



CD4+ cell count < 200/mm³

Pathogen	Presentation	Findings
Histoplasma capsulatum	Fever, weight loss, fatigue, cough, dyspnea, nausea, vomiting, diarrhea	Oval yeast cells with macrophages
HIV	Dementia, HIV-associated neuropathy	Cerebral atrophy on neuroimaging
JC virus (reactivation)	Progressive multifocal leukoencephalopathy	Non-enhancing areas of demyelination on MRI
HHV-8	Kaposi sarcoma, disseminated disease (pulmonary, GI, lymphatic)	
Pneumocystis jirovecii	*Pneumocystis* pneumonia	"Ground-glass" opacities on chest imaging

Figure 10 Part 2



CD4+ cell count < 100/mm³

Pathogen	Presentation	Findings
Bartonella spp	Bacillary angiomatosis	Multiple red to purple papules or nodules, Biopsy with neutrophilic inflammation
Candida albicans	Esophagitis	White plaques on endoscopy; yeast and pseudohyphae on biopsy
CMV	**C**olitis, **R**etinitis, **E**sophagitis,	Linear ulcers on endoscopy, cotton-wool spots on fundoscopy; Biopsy reveals cells lear (owl eye) inclusion bodies
Cryptococcus neoformans	Meningitis	ated yeast on India ink stain or capsular antigen
Cryptosporidium spp	Chronic, watery diarrhea	Acid-fast oocysts in stool
EBV	B-cell lymphoma (e.g. non-Hodgkin lymphoma, CNS lymphoma)	CNA lymphoma-ring enhancing, may be solitary (vs *Toxoplasma*)
Mycobacterium avium-intracellulare, Mycobacterium avium complex	Nonspecific systemic symptoms (fever, night sweats, weight loss) or focal lymphadenitis	Most common if CD4+ cell count < 50/mm³
Toxoplasma gondii	Brain abscesses	Multiple ring-enhancing lesions on MRI

Figure 10 Part 3

Some of these infections have already been covered (EBV, CMV). Others will be covered in future bricks and systems. This is a good resource for future studies, but don't stress on memorizing these until we've covered the specific pathogen.

AIDS-defining conditions are extensive, including over two dozen infections and cancer.

Note that I am including the link to the CDC only for reference, and I do not expect you to know the list of these diseases. **However, know that a patient can have AIDS even if their T cell count is above 200/uL and have one of these conditions.**

In an HIV+ patient iagnosis is made?

How do you test for HIV? (LO4)

There are several tests available for HIV. Some are at-home, self-tests and others are for healthcare professionals. Self-tests may be available for free or a discounted price. However, self-tests are not a substitute for tests administered by health professionals. In the event that a patient is positive on a self-test, a follow-up test must be performed to confirm the results. We will cover the two key tests used in the clinic in this brick.

- **Nucleic Acid Amplification Test (NAAT or NAT)**
- **Antigen/Antibody Test**

In order to determine which and when each test is appropriate, we must revisit the replication cycle of HIV and when host antibodies are produced against the HIV virus after infection.

Observe the following figure:

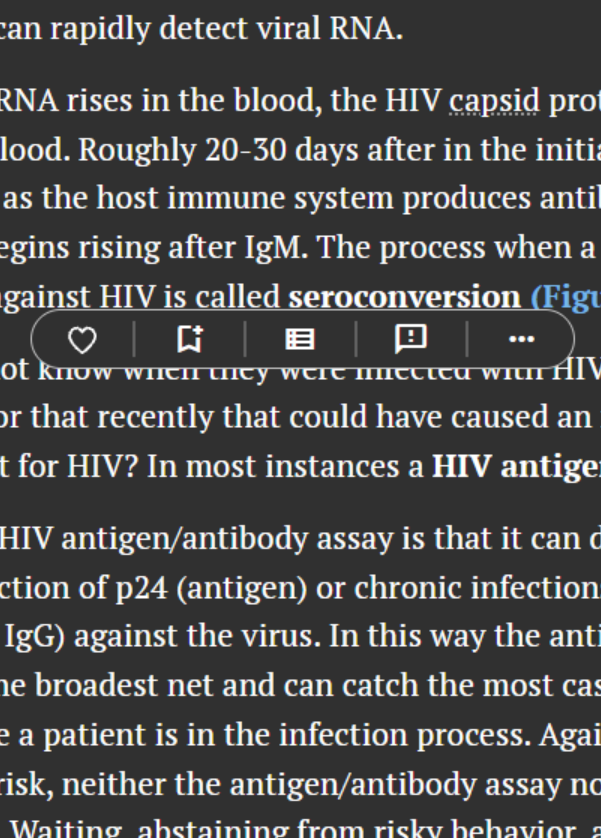


Figure 11: HIV infection progression along with seroconversion

Note that early in an infection, less than 10 days, there is no test that can detect HIV consistently. This is because the virus has not replicated and spread enough to be detected. Patients who have had a potential recent exposure should be advised to return to perform the test and avoid any risky behavior in the interim (Figure 11).

Notice that after 10 days, viral RNA rises rapidly. The earliest test to detect an HIV infection is **NAT (RT-PCR).** Since viral RNA is the first detectable sign of an infection, NAT can rapidly detect viral RNA.

Shortly after viral RNA rises in the blood, the HIV capsid protein p24 becomes detectable in the blood. Roughly 20-30 days after in the initial infection, IgM titers begin to rise as the host immune system produces antibodies against the virus. IgG slowly begins rising after IgM. The process when a patient begins to create antibodies against HIV is called **seroconversion** (Figure 11).

Most patients do not know when they were infected with HIV and might have engaged in behavior that recently that could have caused an infection, so what's the best test for HIV? In most instances a **HIV antigen/antibody assay.**

The benefit of the HIV antigen/antibody assay is that it can detect early infections via detection of p24 (antigen) or chronic infections by detecting host antibodies (IgM or IgG) against the virus. In this way the antigen/antibody assays throw out the broadest net and can catch the most cases of HIV regardless of where a patient is in the infection process. Again, if a patient has a recent exposure risk, neither the antigen/antibody assay nor NAT will be able to detect the virus. Waiting, abstaining from risky behavior, and returning for a follow-up test would be the most appropriate course of action.

CLINICAL CORRELATION

While the antigen/antibody assay catches the majority of HIV cases, it is not always useful. For example, newborns birthed by an HIV+ person should not be screened by the antigen/antibody assay. IgG can cross the placental and cause a false positive antigen/antibody HIV screen for the baby.

All babies born to people with HIV should be screened with NAT at 14 to 21 days of life, at 1 to 2 months, and again at 4 to 6 months.

Place the following in order from when they are detectable in a HIV+ patient's blood, from earliest to latest: IgG, IgM, RNA, p24.

HIV testing algorithm

Because no test has 100% sensitivity/specificity, to diagnose HIV requires two positive tests to occur. As mentioned above, the first is typically the antigen/antibody assay. The second is another immunoassay that detects host antibodies against the HIV-1 and/or HIV-2. While rare, co-infection of HIV-1 and HIV-2 is possible.

If both tests are positive, the patient is informed they have HIV. Follow-up lab tests that establish viral titer in the blood, CD4 count (if not done already), and viral resistance cultures are common tests to perform in patients with HIV.

Here's a summary of a typical HIV testing process:

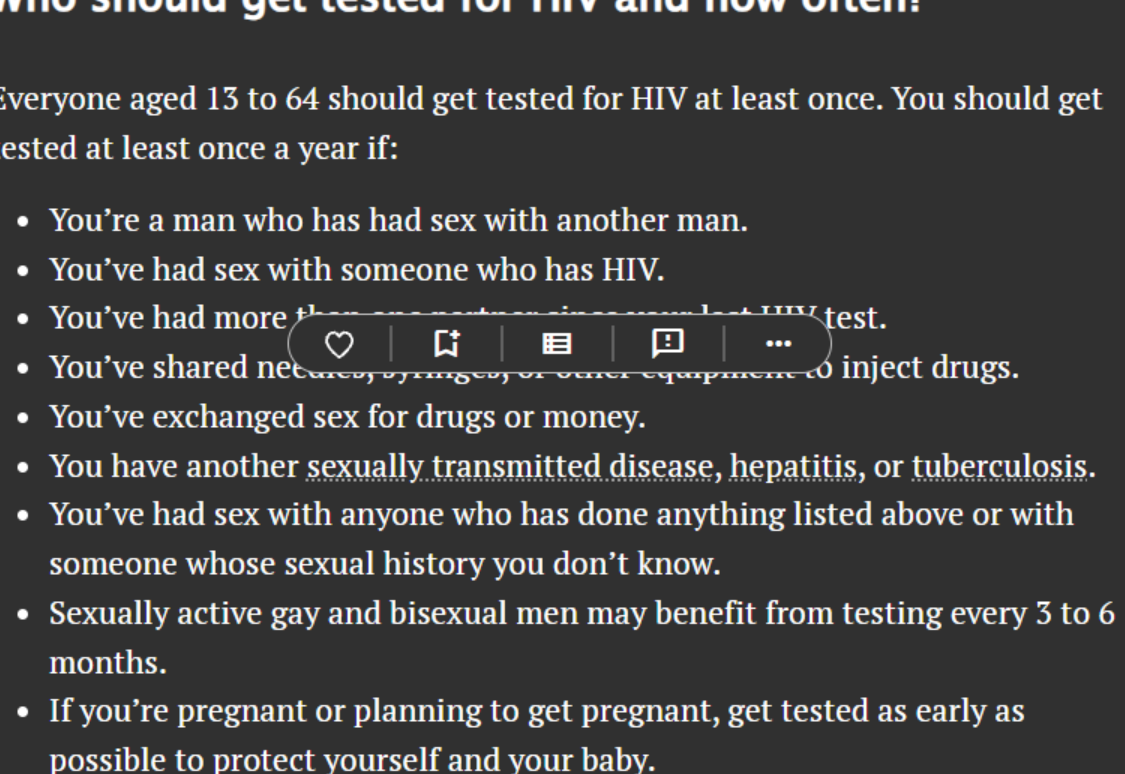

HIV-1/2 Antigen/Antibody Immunoassay
(+)
(-)
Negative for HIV-1 and HIV-2 p24 Ag
HIV-1/HIV-2 Ab Differentiation Immunoassay
HIV-1 (+)
HIV-2 (-)
HIV-1 antibodies detected
HIV-1 (-)
HIV-2 (+)
HIV-2 antibodies detected
HIV-1 (+)
HIV-2 (+)
HIV antibodies detected
HIV-1 (-) or Indeterminate
HIV-2 (-)
HIV-1 NAT
HIV-1 NAT (+)
Acute HIV-1 infection
HIV-1 NAT (-)
Negative for HIV-1

Figure 12: HIV diagnosis flow w/ workup

Once baseline viral titers are established, this allows clinicians to monitor the efficacy of ART. A rise in viral titers while on ART indicates either non-compliance with treatment or that the virus has mutated around the therapy.

Due to HIV's rapid mutation rate, a patient's viral titers must be periodically checked to ensure the therapy is still working. ART therapy has extended the lifespan of HIV patients, but they must have continuity of care to monitor the continued efficacy of the treatment and adjust if mutations occur.

Who should get tested for HIV and how often?

Everyone aged 13 to 64 should get tested for HIV at least once. You should get tested at least once a year if:

- You're a man who has had sex with another man.
- You've had sex with someone who has HIV.
- You've had more test.
- You've shared ne o inject drugs.
- You've exchanged sex for drugs or money.
- You have another sexually transmitted disease, hepatitis, or tuberculosis.
- You've had sex with anyone who has done anything listed above or with someone whose sexual history you don't know.
- Sexually active gay and bisexual men may benefit from testing every 3 to 6 months.
- If you're pregnant or planning to get pregnant, get tested as early as possible to protect yourself and your baby.

CASE CONNECTION

BACK TO INTRODUCTION ↑

The patient most likely has hairy leukoplakia, which is caused by EMV. The lesion cannot be scraped off, which distinguishes it from thrush. However, hairy leukoplakia is commonly seen in immunocompromised patients, so a CBC and HIV antigen/antibody assay is ordered.

The HIV test comes back positive, and the follow-up, confirmatory test is positive for HIV-1 antibodies. The patients T cell count is 372 cells/uL (Normal: 500-1500 cells/uL). The patient in informed and started on ART. His T cell count improves, and his oral lesions resolve.

Summary

- HIV infection is difficult to cure, in part, due to two key steps in its replication pathway: the lack of proofreading by reverse transcriptase, leading to rapid rporation of viral DNA into the host genome.
- The high mutation rate of HIV allows it to quickly adapt to treatments, necessitating the use of multiple drugs simultaneously to increase the number of mutations needed for resistance.
- Despite antiretroviral therapy (ART) suppressing viral titers in the blood below detection, the virus remains in the patient's DNA. ART doesn't remove HIV; it only suppresses viral titers.
- An HIV infection progresses in three phases: acute phase with rapid viral replication, chronic phase with active replication despite no symptoms, and AIDS.
- A diagnosis of AIDS requires either a T cell count of <200/mL or an AIDS-defining illness.
- The antigen/antibody combo assay is the most common screening assay for HIV. It detects both the presence of p24 and patient production of IgM or IgG against the virus.
- Seroconversion is the process when a patient begins producing antibodies against HIV.
- Everyone aged 13 to 64 should get tested for HIV at least once, with higher risk individuals requiring more frequent testing.

Review Questions

1. As part of a phase I clinical trial, a novel HIV drug is utilized to inhibit viral DNA incorporation into the host genome. Patients in the trial noticed a 2-fold greater reduction in blood HIV titers compared to the control. Which protein does this novel drug most likely target?

A. ○ Reverse t
B. ○ gp120/gp41
C. ● Integrase
D. ○ Protease

Hide Explanation

C) Integrase incorporates HIV DNA into host DNA. A) Reverse transcriptase coverts viral RNA into DNA. B) gp120/gp41 facilitate viral entry into host cells by binding CD4 and a co-receptor (CXCR4 OR CCR5). D) HIV protease cleaves the polyprotein into functional proteins after translation of viral DNA.

2. In which patient would NAT be more appropriate over the antigen/antibody assay as a screen for HIV?

A. ○ A patient who had an exposure risk 5 years ago
B. ● A newborn birthed by an HIV+ person
C. ○ A patient who potentially contracted HIV 5 days ago by sharing needles
D. ○ A patient at high risk for HIV with mononucleosis-like symptoms

Hide Explanation

B) is correct. Transplacental IgG antibodies from the pregnant person with HIV infection will cause a false positive in a newborn screened by the antigen/antibody assay, even if the newborn is HIV negative. A) is incorrect since this patient's exposure was 5 years ago. If HIV+, they will be in the chronic phase of illness and the antigen/antibody assay would work just fine. C) is incorrect. No test can detect HIV this early in the infection. The patient should be advised to abstain from risky behavior and come back for further testing. D) This patient is most likely in the acute stage of illness and the antigen/antibody test would work as a screening assay.

How would you rate this Brick?

References

- Appendix A </P>AIDS-Defining Conditions</FONT> (cdc.gov)
- Preventing Perinatal Transmission of HIV After Birth | NIH
- Dating the Origin of the CCR5-Δ32 AIDS-Resistance Allele by the Coalescence of Haplotypes: The American Journal of Human Genetics (cell.com)

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain disease progression and epidemiology of HIV.~~
02 ☑ ~~Describe the life cycle of HIV.~~
03 ☑ ~~Interpret laboratory results to identify HIV.~~

Objective Confidence: 3/3